Day-4

HTML5 Design

1. XHTML vs. HTML5

XHTML (Extensible Hypertext Markup Language) and **HTML5** are both markup languages used for structuring web content. However, they differ in several ways:

- **XHTML** is stricter in syntax than HTML5. It requires all tags to be closed, attributes to be in lowercase, and elements to be properly nested.
- **HTML5** is more lenient with syntax. It supports unclosed tags, case-insensitive attributes, and more flexible nesting. HTML5 also introduces new elements and attributes for a richer semantic structure.

Example: XHTML vs. HTML5

XHTML Example:

```html
<!DOCTYPE html PUBLIC "-//W3C//DTD XHTML 1.0 Strict//EN"
  "http://www.w3.org/TR/xhtml1/DTD/xhtml1-strict.dtd">
<html xmlns="http://www.w3.org/1999/xhtml">
  <head>
    <title>XHTML Example</title>
  </head>
  <body>
    <p>This is an example of XHTML.</p>
    <img src="tree.jpg" alt="Tree" />
  </body>
</html>
```

HTML5 Example:

```html
<!DOCTYPE html>
<html lang="en">
  <head>
    <meta charset="UTF-8">
    <title>HTML5 Example</title>
  </head>
```

```html
  <body>
    <p>This is an example of HTML5.</p>
    <img src="tree.jpg" alt="Tree">
  </body>
</html>
```

2. Building with HTML5

HTML5 introduces new elements and attributes that provide better structure and semantics for web documents. These include:

- **Semantic Elements**: **<header>, <footer>, <article>, <section>, <nav>, <aside>, and <main>.**
- **Form Elements**: **<input type="email">, <input type="date">, <input type="range">**, etc., which provide better input types and validation.
- **Multimedia**: **<audio>, <video>, and <canvas>** elements for multimedia content and graphics.

Example: Basic HTML5 Structure

```html
<!DOCTYPE html>
<html lang="en">
  <head>
    <meta charset="UTF-8">
    <meta name="viewport" content="width=device-width, initial-scale=1.0">
    <title>HTML5 Structure</title>
  </head>
  <body>
    <header>
      <h1>Welcome to HTML5</h1>
    </header>
    <nav>
      <ul>
        <li><a href="#home">Home</a></li>
        <li><a href="#about">About</a></li>
        <li><a href="#contact">Contact</a></li>
      </ul>
    </nav>
    <main>
      <article>
        <h2>Main Content Area</h2>
        <p>This is the main content area.</p>
      </article>
    </main>
```

```html
    <footer>
      <p>&copy; 2024 HTML5 Example. All rights reserved.</p>
    </footer>
  </body>
</html>
```

3. HTML5 Audio & Video

HTML5 provides native support for audio and video elements, allowing developers to embed multimedia content without relying on third-party plugins like Flash.

Example: HTML5 Audio and Video

```html
<!DOCTYPE html>
<html lang="en">
  <head>
    <meta charset="UTF-8">
    <title>HTML5 Audio & Video Example</title>
  </head>
  <body>
    <!-- Audio Element -->
    <audio controls>
      <source src="audio-file.mp3" type="audio/mpeg">
      Your browser does not support the audio element.
    </audio>
    <br><br>

    <!-- Video Element -->
    <video width="320" height="240" controls>
      <source src="video-file.mp4" type="video/mp4">
      Your browser does not support the video element.
    </video>
  </body>
</html>
```

4. HTML5 Video Captions

HTML5 supports captions and subtitles for videos using the <track> element within the <video> tag.

Example: HTML5 Video with Captions

```html
<!DOCTYPE html>
<html lang="en">
  <head>
```

```html
    <meta charset="UTF-8">
    <title>HTML5 Video with Captions</title>
  </head>
  <body>
    <!-- Video with Captions -->
    <video width="320" height="240" controls>
      <source src="video-file.mp4" type="video/mp4">
      <track src="subtitles_en.vtt" kind="subtitles" srclang="en"
label="English">
      Your browser does not support the video element.
    </video>
  </body>
</html>
```

Explanation of <track> Attributes:

- **src**: Specifies the URL of the subtitle file.
- **kind**: Specifies the type of text track (e.g., subtitles, captions, descriptions).
- **srclang**: Specifies the language of the track text data (e.g., "en" for English).
- **label**: Provides a user-readable title for the track.

Styling With CSS

1. Overview of CSS

CSS (Cascading Style Sheets) is used to control the presentation of HTML documents, including layout, colors, fonts, and more. CSS allows for separation of content (HTML) and design, making it easier to maintain and update web pages.

2. Adding CSS and How

CSS can be added to HTML documents in three ways:

1. **Inline CSS**: Using the style attribute directly in HTML tags.
2. **Internal CSS**: Placing CSS code within a **<style>** tag in the **<head>** section of an HTML document.
3. **External CSS**: Linking to an external **.css** file using the <link> tag.

Example: Adding CSS

```html
<!DOCTYPE html>
<html lang="en">
  <head>
    <meta charset="UTF-8">
    <title>Adding CSS Example</title>
    <!-- Internal CSS -->
    <style>
      body {
        font-family: Arial, sans-serif;
        background-color: #f0f0f0;
        color: #333;
      }
      .container {
        width: 80%;
        margin: auto;
        padding: 20px;
        background-color: white;
        border-radius: 10px;
        box-shadow: 0 0 10px rgba(0, 0, 0, 0.1);
      }
    </style>
    <!-- External CSS -->
    <link rel="stylesheet" href="style.css">
  </head>
  <body>
    <div class="container">
      <h1>Welcome to CSS Styling</h1>
      <p>This is an example of how to add CSS to your HTML.</p>
    </div>
  </body>
</html>
```

3. Fonts

CSS allows you to specify fonts for your web content using properties like font-family, font-size, font-weight, and font-style.

Example: Using Different Fonts

```html
<!DOCTYPE html>
<html lang="en">
  <head>
    <meta charset="UTF-8">
    <title>Font Styling Example</title>
    <style>
      h1 {
        font-family: 'Arial', sans-serif;
```

```
        font-size: 36px;
        font-weight: bold;
      }
      p {
        font-family: 'Georgia', serif;
        font-size: 16px;
        font-style: italic;
      }
    </style>
  </head>
  <body>
    <h1>CSS Fonts Example</h1>
    <p>This paragraph uses Georgia font with italic style.</p>
  </body>
</html>
```

4. Colors

CSS allows you to apply **colors** to text, backgrounds, borders, and other elements using color names, hexadecimal values, **RGB, RGBA, HSL, and HSLA** color models.

Example: Applying Colors

```
<!DOCTYPE html>
<html lang="en">
  <head>
    <meta charset="UTF-8">
    <title>Color Styling Example</title>
    <style>
      body {
        background-color: #f0f0f0;
        color: #333333;
      }
      .highlight {
        color: #ff6600; /* Hexadecimal color */
        background-color: rgba(255, 255, 0, 0.3); /* RGBA color with opacity
*/
      }
    </style>
  </head>
  <body>
    <p>This is a standard paragraph.</p>
    <p class="highlight">This paragraph is highlighted with color.</p>
  </body>
</html>
```

5. Shadowing

CSS provides two properties for applying shadows to elements:

- **Text Shadow (text-shadow)**: Adds shadow to text.
- **Box Shadow (box-shadow)**: Adds shadow to elements like divs, buttons, etc.

Example: Text and Box Shadows

```
<!DOCTYPE html>
<html lang="en">
  <head>
    <meta charset="UTF-8">
    <title>Shadow Example</title>
    <style>
      h1 {
        text-shadow: 2px 2px 5px #000;
      }
      .box {
        width: 200px;
        height: 100px;
        background-color: #f0f0f0;
        box-shadow: 5px 5px 15px rgba(0, 0, 0, 0.3);
        margin: 20px;
        padding: 20px;
      }
    </style>
  </head>
  <body>
    <h1>Text Shadow Example</h1>
    <div class="box">Box Shadow Example</div>
  </body>
</html>
```

6. Rounded Corners

CSS3 introduced the border-radius property to create rounded corners on elements.

Example: Rounded Corners

```
<!DOCTYPE html>
```

```html
<html lang="en">
  <head>
    <meta charset="UTF-8">
    <title>Rounded Corners Example</title>
    <style>
      .rounded {
        width: 200px;
        height: 100px;
        background-color: #3498db;
        border-radius: 15px;
        margin: 20px;
        padding: 20px;
        color: white;
        text-align: center;
        line-height: 60px;
      }
    </style>
  </head>
  <body>
    <div class="rounded">Rounded Corners</div>
  </body>
</html>
```

7. Gradients

CSS3 gradients allow for smooth transitions between multiple colors. There are two types: **linear gradients** and **radial gradients**.

Example: CSS Gradients

```html
<!DOCTYPE html>
<html lang="en">
  <head>
    <meta charset="UTF-8">
    <title>Gradient Example</title>
    <style>
      .gradient-box {
        width: 200px;
        height: 200px;
        background: linear-gradient(to right, #ff7e5f, #feb47b);
        margin: 20px;
      }

      .radial-gradient-box {
        width: 200px;
        height: 200px;
        background: radial-gradient(circle, #ff7e5f, #feb47b);
```

```
      margin: 20px;
    }
  </style>
</head>
<body>
  <div class="gradient-box">Linear Gradient</div>
  <div class="radial-gradient-box">Radial Gradient</div>
</body>
</html>
```

These examples provide a basic overview of HTML5 and CSS3 features, showcasing how they can be used to enhance web pages. Copy the examples into your text editor and open them in a web browser to see them in action.